

Abstract

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1 Introduction

1.1 Context and Motivation

1.2 Research Objectives

1.3 Main Contributions of the Thesis

1.4 Structure of the Document

2 Theoretical Background

The field of artificial intelligence has undergone deep transformations since its inception, evolving through distinct paradigms that reflect different philosophical and technical approaches to machine intelligence. This chapter provides the theoretical foundations necessary to understand the framework proposed in this thesis, exploring the evolution of learning paradigms in AI, some techniques of visual perception and the principles underlying symbolic reasoning and reinforcement learning.

2.1 The Evolution of Learning in Artificial Intelligence

2.1.1 Artificial Intelligence Paradigms

The history of artificial intelligence is characterized by the emergence and evolution of three major paradigms, each representing a distinct approach to providing machines the human-like reasoning abilities. The symbolic paradigm, dominant from the 1950s through the 1980s. This paradigm was grounded in the hypothesis that human intelligence could be replicated through the manipulation of symbols, according to logical rules, to perform reasoning tasks. The systems built on this paradigm operated on explicit knowledge representations, using formal logic and rule-based reasoning to solve problems. These systems are grounded in formal logic and manually designed rule sets. Classic examples include expert systems such as MYCIN and Deep Blue, as well as semantic networks and logic programming frameworks [1]. They are capable of performing reasoning tasks and have achieved notable success in constrained domains where knowledge can be explicitly encoded. However, the symbolic paradigm faced significant limitations due to its high reliance on manually encoded knowledge. Indeed, when faced with tasks requiring pattern recognition, learning from experience, the management of uncertainty or incomplete information, it suffered from brittle-

ness [1, 2]. These challenges gave rise to the connectionist paradigm, which gained prominence in the late 1980s and experienced a resurgence in the 2010s with the advent of deep learning [3]. Inspired by the structure and function of biological neural networks, connectionist systems learn representations directly from data through training processes that adjust connection weights between artificial neurons [3, 4]. This paradigm has achieved remarkable success in tasks such as image recognition, natural language processing and game playing, often surpassing human-level performance in specific domains [5]. Their capacity for robust and interpretable reasoning remains limited. Deep learning still struggles with structured reasoning, causal inference and factual consistency. Despite the impressive capabilities of neural networks, their operation and internal representation often lacks transparency, giving rise to what is commonly referred to as the "black box problem" [2], which limits interpretability, accountability and trust. Indeed, many of these approaches compromised interpretability and relied on strong assumptions. Moreover, they lacked the rich semantic expressiveness characteristic of symbolic systems. These limitations provide strong motivation for integrating neural and symbolic paradigms. In short, symbolic systems excel at reasoning but lack at perception, while neural networks excel at perception but struggle with reasoning [1]. This limitation has motivated the emergence of a hybrid paradigm, which seeks to combine the learning capabilities of neural networks with the interpretability and reasoning power of symbolic systems. Neuro-symbolic AI represents this third wave, aiming to bridge the gap between data-driven learning and knowledge-based reasoning as well as between logic-based inference and gradient-based learning. By integrating these paradigms, it aims to potentially offer systems that are both powerful and explainable [6]. Neuro-symbolic methods have demonstrated significant promise across a wide range of application areas, especially in scenarios that demand the combination of data-driven learning and structured symbolic reasoning [6]. This emerging paradigm reflects a synthesis of complementary approaches, bringing together the statistical strengths of deep learning, the formal foundations of symbolic logic and the adaptability of large-scale pretrained models.

Nonetheless, several important challenges remain. These include:

- Ensuring compatibility between the discrete symbolic structures used in symbolic approaches and the continuous vector representations typical of neural networks.
- Enabling the model to dynamically learn, adapt, modify, and refine logical rules.

- Achieving an optimal balance between system complexity and computational efficiency.
- Addressing the absence of unified architectural frameworks by developing a general architecture that systematically and reusably integrates neural and symbolic components.

Although several challenges persist, neuro-symbolic AI is widely regarded as a promising direction for developing systems capable of transparent, reliable and generalizable reasoning.

2.1.2 Symbolic and Hybrid Approaches in Artificial Intelligence

Symbolic AI operates on the principle that intelligence can be reduced to the manipulation of symbols representing concepts, objects and relationships. The knowledge in symbolic systems is typically encoded using various representation schemes, including production rules, semantic networks, frames and logical predicates. These representations enable explicit reasoning processes such as deduction, induction and abduction, allowing systems to draw inferences, explain their decisions and incorporate domain knowledge provided by human experts. The advantages of symbolic approaches include interpretability, the ability to incorporate prior knowledge and support for logical reasoning and explanation. However, they also present significant challenges: knowledge acquisition bottleneck, brittleness when facing situations not covered by explicit rule and difficulty in learning from raw sensory data. The manual encoding of knowledge is intensive and may not capture the full complexity of real-world domains [1]. Hybrid approaches seek to overcome these limitations by integrating symbolic and connectionist methods. Various integration strategies have been proposed, including systems where neural networks generate symbolic representations that are then processed by reasoning engines, architectures that embed symbolic knowledge into neural network structures and frameworks that use symbolic reasoning to guide neural learning processes [6]. Recent work in neuro-symbolic AI has demonstrated promising results in domains requiring both pattern recognition and logical reasoning, such as visual question answering, knowledge graph completion, and program synthesis. Following the advent of neuro-symbolic artificial intelligence, a paradigm shift has been induced in the way machines learn. This approach combines deductive, rule-based learning, typical of symbolic artificial intelligence, with the inductive and pattern-recognition capabilities characteristic of neural networks. The result is a

hybrid approach that leverages the strengths of both domains, leading to a more comprehensive learning methodology. Neuro-symbolic concepts demonstrate strong compositional generalization: new concepts can be created by structurally combining previously learned or existing ones [7]. This compositionality naturally allows for a decomposition of the learning problem, in order to learn and recognize basic concepts. During the inference phase, these concepts can be recombined to achieve the objectives of more complex tasks. Neuro-symbolic AI is one of the most exciting directions in AI research today. It aims to create intelligent systems capable of learning and reasoning like humans.

2.1.3 Reinforcement Learning: Learning Through Experience

Reinforcement learning (RL) represents a distinct learning paradigm in which an agent learns to make decisions by interacting with an environment. Unlike supervised learning, which requires labeled training data, or unsupervised learning, which seeks to discover patterns in unlabeled data, reinforcement learning is based on the principle of learning from the consequences of actions through trial and error [8]. The agent takes actions, observes their consequences, and receives feedback in the form of rewards. At the beginning of learning, an agent may have limited or no prior knowledge of the environment. In model-free approaches, no explicit model is available, and learning relies entirely on interaction. In contrast, model-based methods assume a known model or aim to learn it explicitly. This flexibility makes reinforcement learning suitable for complex environments, where providing a complete dataset or manually specifying optimal behavior is impractical. The core structure of Reinforcement Learning (RL) is modelled through a Markov Decision Process (MDP), which provides a mathematical framework for sequential decision-making [8]. An MDP is defined by a finite or continuous set of states representing the possible situations an agent can encounter, a set of actions available to the agent and a transition function that specifies the probability of moving from one state to another after taking a particular action. Additionally, a reward function assigns numerical feedback to each state-action pair, quantifying the immediate benefit of a given choice. By observing the current state, selecting an action, and receiving a reward and a new state, the agent incrementally improves its decision-making strategy [9]. Over time, the agent aims to learn an optimal policy, which is a strategy mapping states to actions. The goal of this policy is to maximize the expected cumulative reward across future interactions, balancing short-term

gains with long-term benefits through exploration and exploitation. Several agent design approaches exist, including methods based on value estimation and approaches that learn action values directly, such as Q-learning. More advanced techniques combine different strategies to improve learning efficiency. Due to its ability to learn from experience, reinforcement learning has achieved notable success in domains such as game playing, robotics, and resource management [9].

2.2 Computer Vision Techniques

2.2.1 Classical Techniques for Video Analysis

2.2.2 Visual Perception in Dynamic Environments

2.2.3 Deep Learning for Feature Extraction from Video Streams

2.3 Knowledge Representation and Reasoning

2.3.1 Symbolic Representations, Semantic Networks, and Interpretable Models

2.4 Reinforcement Learning Foundations

2.4.1 Reinforcement Learning Principles and Deep Q-Networks

2.5 Atari-like Environments as Testbeds

2.5.1 Characteristics and Challenges of 2D Game Environments

3 State of the Art and Challenges

3.1 Challenges in Symbolic Interpretation

3.1.1 The Illusion of Understanding

The Lack of World Models

3.1.2 The Black Box Problem

Making Sense of AI: Toward Explainability

3.1.3 Learning Paradigms and Explainable AI

Learning Without Understanding

Explainable AI: Goals and Approaches

Post-hoc Explainability

Intrinsic Interpretability

Why XAI is Not Enough

3.2 Cognitive Foundations

3.2.1 Human World Modelling and the Origins of Structured Cognition

3.2.2 Core Knowledge Theory: Cognitive Principles and Psychological Foundations

3.2.3 Definition and Relevance for World Understanding

3.2.4 Cognitive Models and Symbolic Representation

3.3 Positioning of the Proposed Framework

4 Framework Design - Proposed Framework

4.1 Conceptual Foundations

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4.2.4 Generic Environment as a Transitional Representation

4.2.5 Deep Q-Network in a Domain-Agnostic Environment

5 Implementation

5.1 End-to-End Pipeline

5.2 General Architecture of the Framework

5.2.1 Design Choices

5.2.2 Selected Techniques and Algorithms

5.2.3 Modular Structure of the System

Video Acquisition and Preprocessing Module

Feature Extraction Module

Symbolic Mapping Module - Symbolic Reasoning Engine

Integration of Core Knowledge for Semantic Understanding

RL Agent

5.2.4 Optimizations and Efficiency Considerations

6 Experimental Evaluation

6.1 Datasets and Testing Scenarios

6.1.1 Evaluation Metrics: Accuracy, Interpretability, Scalability

6.2 Experimental Results

6.2.1 Comparison with Existing Methods

6.2.2 Critical Discussion of the Results

7 Applications and Future Perspectives – Conclusion

7.1 Future Directions

7.1.1 Potential Applications of the Framework

7.1.2 Possible Extensions of the Work

7.1.3 Open Challenges and Future Research Directions

7.2 Summary of Contributions, Impact of the Research, and Final Reflections

References

Bibliography

- [1] B. Liang, Y. Wang, and C. Tong, “Ai reasoning in deep learning era: From symbolic ai to neural-symbolic ai,” *Mathematics*, vol. 13, no. 11, p. 1707, 2025.
- [2] G. Marcus, “Deep learning: A critical appraisal,” *arXiv preprint arXiv:1801.00631*, 2018. [Online]. Available: <https://arxiv.org/abs/1801.00631>
- [3] J. Schmidhuber, “Deep learning in neural networks: An overview,” *Neural Networks*, vol. 61, pp. 85–117, 2015.
- [4] A. S. d’Avila Garcez, L. C. Lamb, and D. M. Gabbay, *Neural-Symbolic Cognitive Reasoning*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009. [Online]. Available: <https://link.springer.com/book/10.1007/978-3-540-73246-4>
- [5] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, pp. 436–444, 2015. [Online]. Available: <https://doi.org/10.1038/nature14539>
- [6] A. S. d’Avila Garcez, T. R. Besold, L. De Raedt, P. Földiak, P. Hitzler, T. Icard, and D. L. Silver, “Neural-symbolic learning and reasoning: Contributions and challenges,” in *AAAI Spring Symposia*, Mar. 2015, pp. 18–21. [Online]. Available: <https://www.aaai.org/Library/Symposia/Spring/ss15-01.php>
- [7] B. M. Lake, T. D. Ullman, J. B. Tenenbaum, and S. J. Gershman, “Building machines that learn and think like people,” *Behavioral and Brain Sciences*, vol. 40, p. e253, 2017.
- [8] L. P. Kaelbling, M. L. Littman, and A. W. Moore, “Reinforcement learning: A survey,” *Journal of Artificial Intelligence Research*, vol. 4, pp. 237–285, 1996.

- [9] M. L. Puterman, *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. Hoboken, NJ, USA: Wiley, 2014.

A Additional Technical Details

B Code Snippets or Pseudocode of Relevant Modules

C Dataset Specifications or Experimental Settings